

RISHABH SHUKLA

Backend & ML Engineer

Chicago, IL · (205) 830-2801 · shuklarishabh487@gmail.com · [LinkedIn](#) · [GitHub](#)

SUMMARY

Backend Software Engineer **3 plus years** of experience with focus on **designing resilient systems** to solve **critical production challenges**, including resolving **database deadlocks** and **transaction collisions**, architecting **high-throughput data pipelines**, and optimizing **CI/CD for reliability**. Adept at enhancing data consistency, system fault tolerance, and deployment velocity leading to significant reductions in operational costs and engineering overhead. Experienced in building **scalable microservices** and **cloud-native applications** that improve performance while maintaining **high availability**. Published **ML researcher (arXiv 2026)** and **Global Rank 19** in **HackerEarth ML Challenge**. Proven track record eliminating production race conditions, cutting infrastructure costs by 18%, and shipping **APIs** that scale to **500K+** monthly requests. Seeking roles where I can apply systems thinking to reduce operational overhead, improve reliability, and deliver measurable business impact.

EXPERIENCE

Backend Developer · Reliance IQ

Sep 2025 – Jan 2026

- Built a **RAG based LLM** powered parts discovery system using **Firecrawl** for web data ingestion, with **Django** and **Python FastAPI** for heavy equipment OEMs (Kubota, Bobcat). Indexed over **200,000 parts** with vendor availability, alternate SKUs, and delivery windows. Reduced procurement lookup time from **15 minutes** of manual spreadsheet work to **under 10 seconds**, saving clients **90 to 110 hours per month** in coordination overhead. This translated to **\$0.77 millions** in annual operational cost avoidance across all clients.
- Ingested a **200,000 row** OEM parts list under a tight **3 day deadline**. Re-architected the upload pipeline using **async** chunked processing with per row schema validation. Cut data rejection rate from **34%** to under **3%** on first run, eliminating brittle manual imports.
- Automated **GCS uploads** using **signed URLs**, replacing a manual handoff process that required **15+ hrs** of monthly engineering time, cutting file management overhead by **60%** and enabling direct uploads without backend involvement.
- Engineered **30 plus REST endpoints** using **Django** and **FastAPI** for parts search and ingestion status, serving a **React frontend**. Added request level validation with **Pydantic** that blocked malformed part numbers before **DB insertion**. Resolved a recurring class of search corruption issues that had been surfacing in production.
- Owned end to end **GCP deployments** as the sole DevOps lead on a **3 person team**. Wrote a **Python based deployment automation** script using **GCP Cloud SDK** that cut release time from **28 minutes** of manual steps to under **4 minutes**. Enabled the team to ship twice as often without working extra hours, saving **\$4,200 per month** in engineering overhead that was being burned on manual release coordination.

Backend Developer · Incedo Inc.

Jan 2020 – Dec 2022

- Designed and owned **70+** client facing **REST APIs** using **Django (Python)** and **Express.js (Node.js)** across **3 product** lines in a **150** engineer org. Optimized **N+1 queries** in **Django ORM** and **Mongoose** populate calls that were causing **p99 latency** spikes above **4 seconds** under peak load, bringing them down to under **400ms** consistently and preventing **\$1.8 million** annually in SLA breach penalties.
- Traced a bug that had been silently corrupting order records for months. Two concurrent POST requests were both passing the existence check before either committed, creating duplicates and missing entries. Identified missing **select_for_update()** lock in the **Django ORM** write path (**Python**) and added pessimistic row locking. Data corruption dropped to **zero** on the next release, saving the team **4 to 6 hours per sprint** that used to go into chasing phantom order issues.
- Wrote **100+** integration test suites using **PyTest (Python)** and **Mocha with Chai (Node.js)** covering order, inventory and billing paths that had historically caused most production incidents. Post deploy bug rate dropped from near weekly to **zero** over the following quarter.
- Collaborated with SRE to retune **Kubernetes** HPA thresholds and resource limits for **10+ high traffic Django** and **Express.js services**. Services stopped getting OOM killed during peak hours. Infra spend dropped by **18%** without any SLA impact, freeing up approx **2.4 million** to **3 million** over three years that got redirected to feature development.
- Set up **Jenkins CI/CD** pipelines replacing manual **SSH deploys** across **12 services**. Removed a whole category of human error. Cut average deploy time from **40 minutes** to **8 minutes**, letting the team ship on demand instead of batching releases every **2 to 3 days**. This recovered **10 to 12** engineering hours per week across the team

EDUCATION

M.S. Computer Science — University of Alabama at Birmingham

Aug 2023 – May 2025

B.Tech Computer Science — Gyan Ganga Institute of Technology & Science

Sep 2016 – May 2020

TECHNICAL SKILLS

Languages: Python, Go, JavaScript, TypeScript, Java, SQL

Backend: Django, Flask, FastAPI, Node.js/Express, Spring Boot, REST, GraphQL, gRPC, Celery, Kafka, RabbitMQ

ML / AI: LLM, RAG, Langchain, TensorFlow, scikit-learn, pandas, NumPy, PyTorch, Computer Vision

Cloud & DevOps: AWS (EC2, ECS, Lambda, S3, DynamoDB, SNS/SQS), GCP, Docker, Kubernetes, Terraform, CI/CD

Databases: PostgreSQL, MySQL, MongoDB, Redis, DynamoDB

PUBLICATIONS & RESEARCH

VGG-UNet Hybrid Architecture for Colorectal Polyp Segmentation · arXiv preprint, 2025 | [GitHub](#)

- Identified that standard **U-Net's** random encoder initialization underperforms on small medical datasets; proposed replacing it with a pretrained **VGG16** backbone, achieving **Dice 0.9297 (+15.54pts over baseline)** on CVC-ClinicDB, competitive with 2024–2025 SOTA methods.
- Designed and ran an 8-experiment ablation study on a 612-image colonoscopy dataset, systematically varying **loss functions (BCE-Dice, Dice, Focal)**, **encoder fine-tuning strategy (frozen vs unfrozen)**, backbone depth (**VGG16 vs VGG19**), and augmentation intensity, producing actionable findings on each variable's impact on segmentation performance.

PROJECTS

SVD-Based Image Forgery Detection Computer Vision · PyTorch | [GitHub](#)

- Designed a forensic detection pipeline that applies **Singular Value Decomposition** to extract **statistical features** from image blocks anomalies in **singular values** reveal tampered regions introduced by splicing or copy-move attacks.
- Fed extracted **SVD** features into a **CNN classifier** achieving **94% detection accuracy**; added **heatmap visualization** to localize tampered regions, making results interpretable for legal and journalistic use cases.

Nextgen Bank Django · FastAPI · Security | [GitHub](#)

- Engineered a secure digital banking platform with **OTP-based authentication**, **bcrypt-hashed password storage**, and **virtual card management** designed to prevent common attack vectors including **credential stuffing** and **session hijacking**.
- Built **RESTful APIs** handling **thousands of daily fund transfers** and **real-time account balance updates**; enforced **transaction atomicity** and **rollback logic** to ensure **zero reconciliation errors** across all test scenarios.

MERN Invoice Generator Full Stack · Docker | [GitHub](#)

- Architected a full-stack invoice and quotation platform with **MongoDB**, **Express**, **React**, and **Node.js**; containerized with **Docker** and **NGINX** as reverse proxy for **consistent local and production environments**.
- Implemented **Redux Toolkit/RTK-Query** for **server state management**, **reducing redundant API calls by 40%** through intelligent **cache invalidation** and **optimized re-fetch** strategies, improving **load time for frequently accessed invoice data**.

ACHIEVEMENTS & CERTIFICATIONS

- HackerEarth ML Challenge** — **Global Rank #19** out of 1000+ participants (May 2021)
- AWS Certified Cloud Practitioner** (Jun 2021)